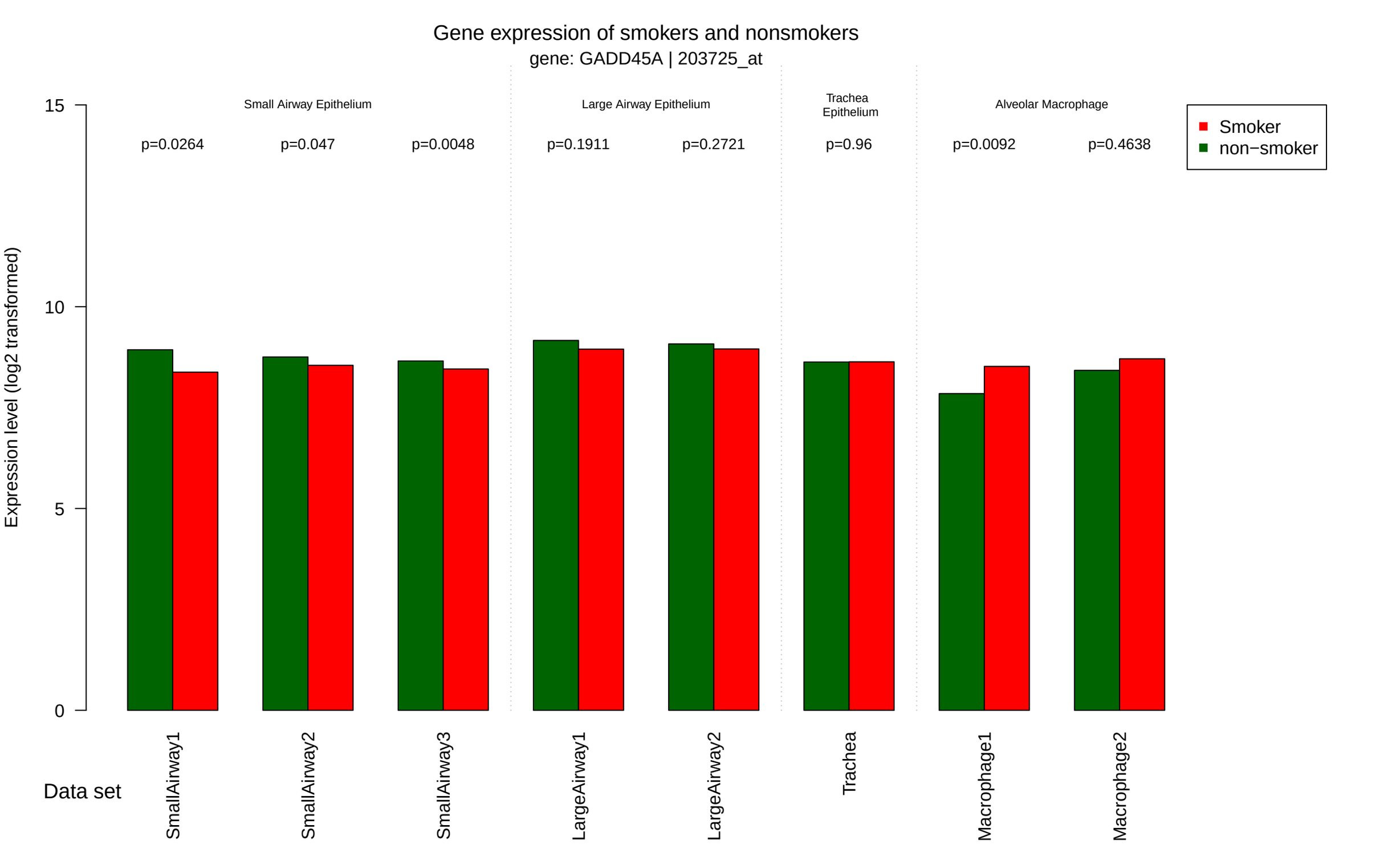
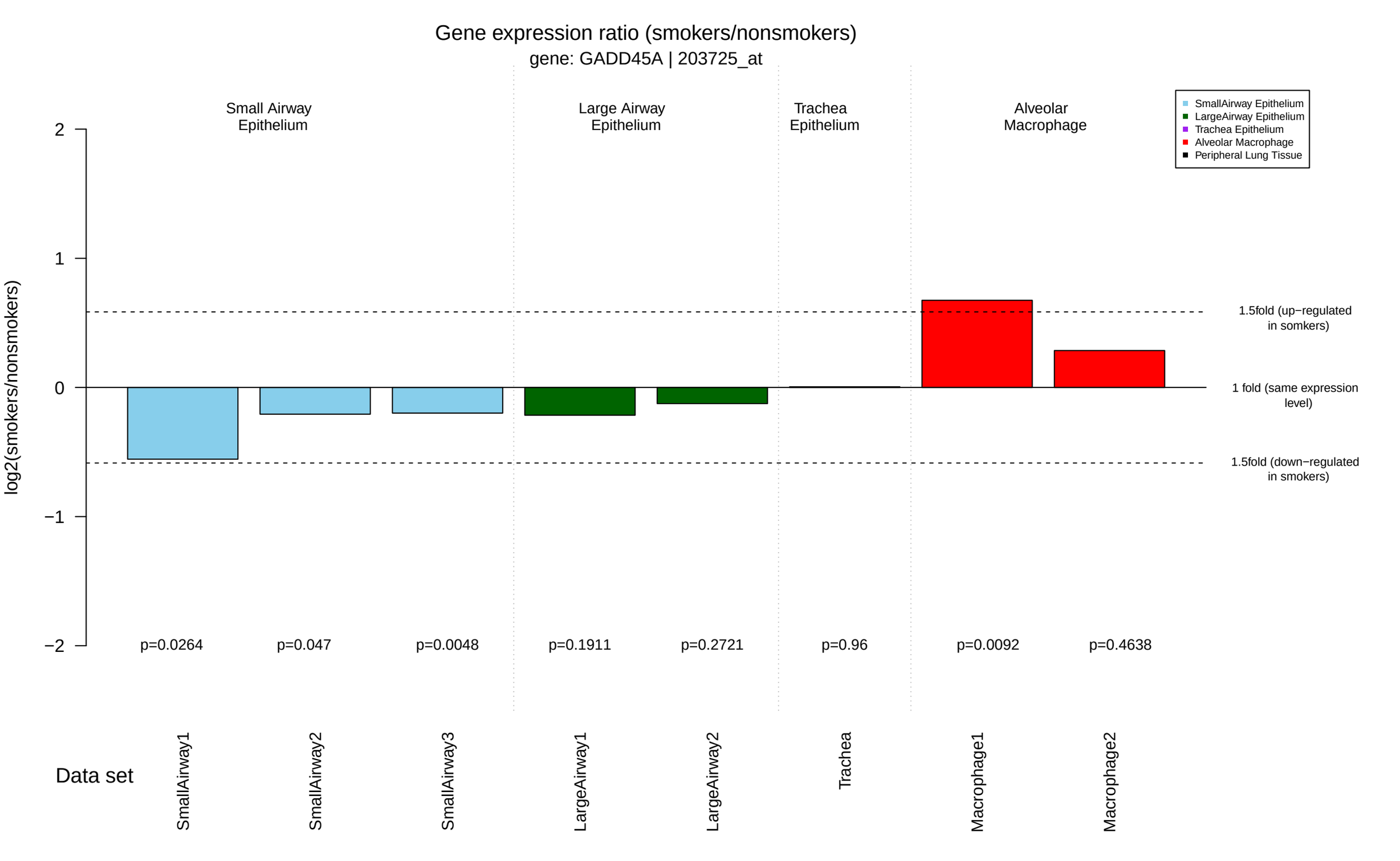
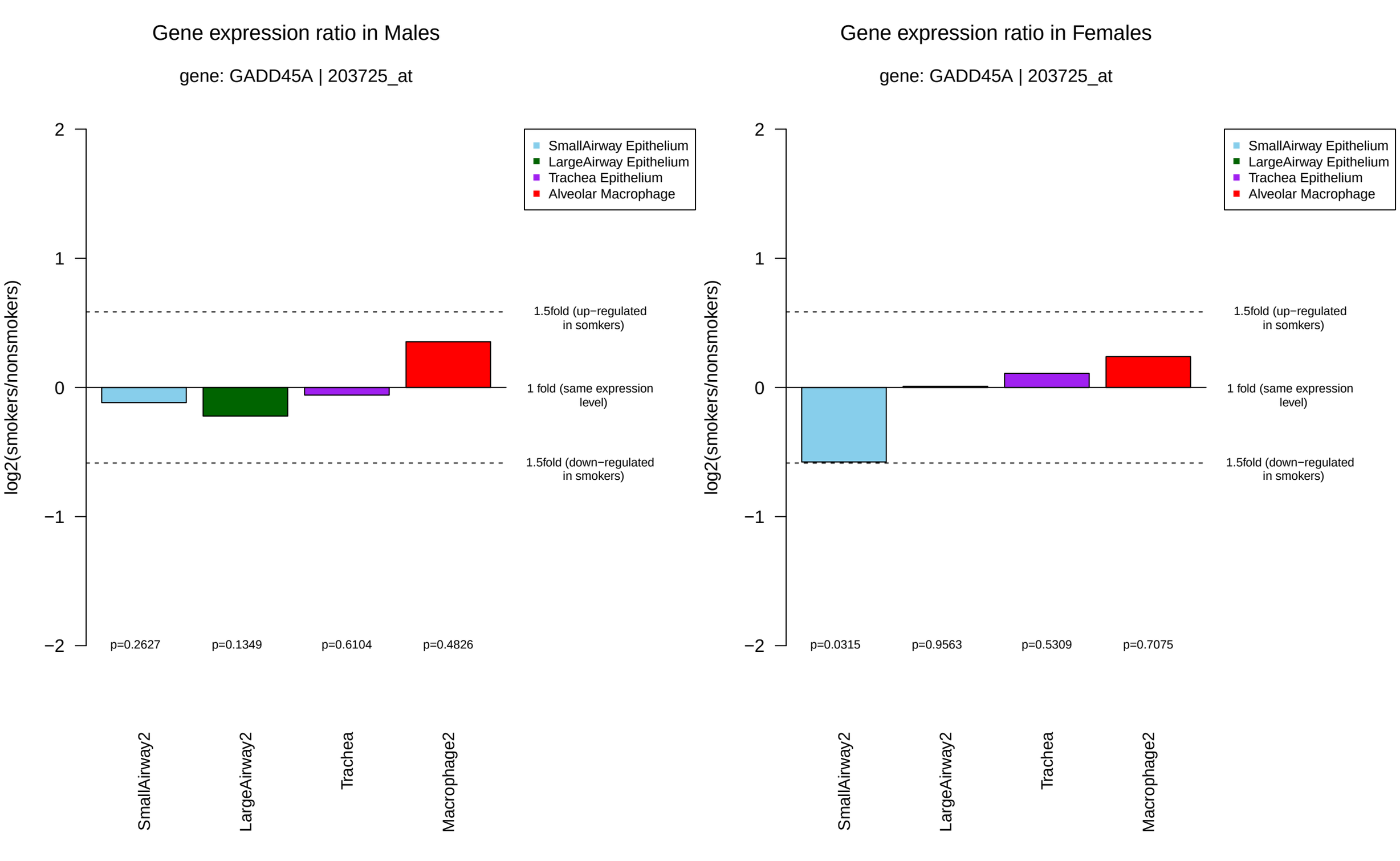
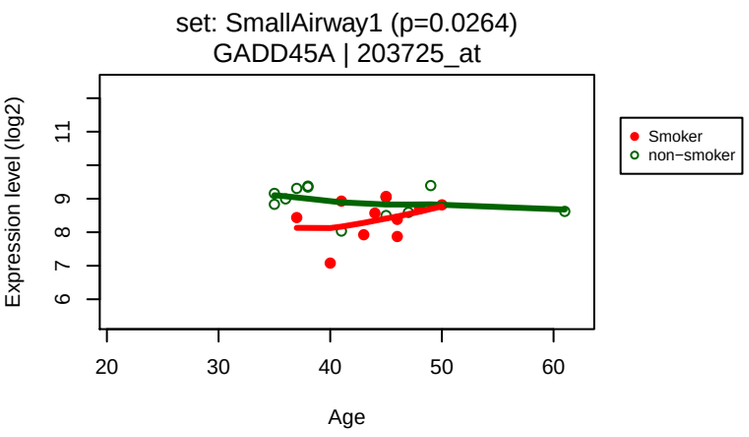
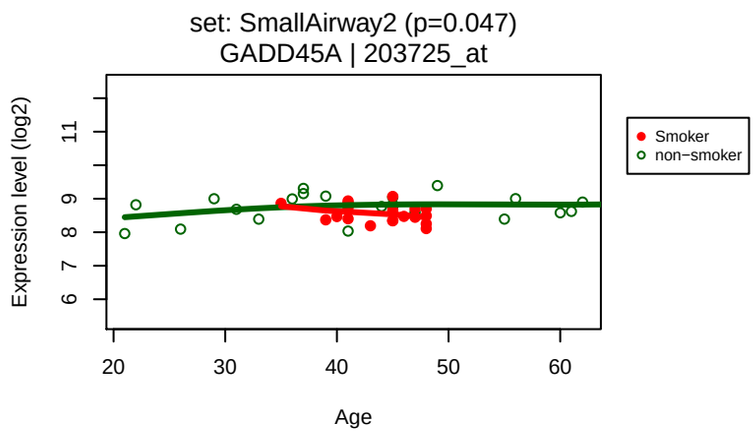
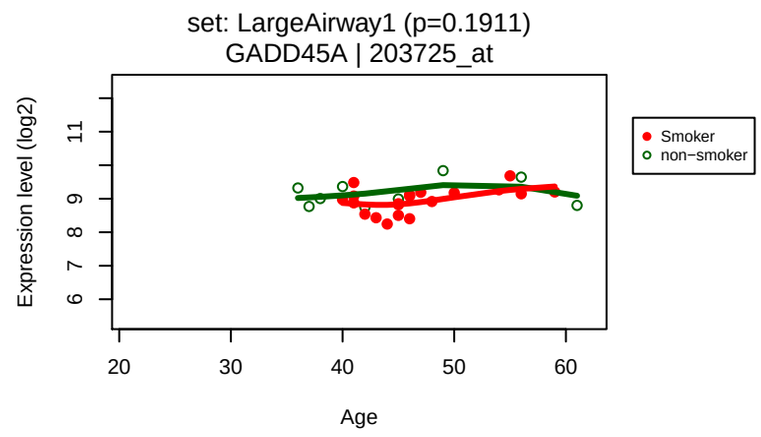

Gene expression of smokers and nonsmokers
gene: GADD45A | 203725_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0264
p=0.047
p=0.0048
p=0.1911
p=0.2721
p=0.96
p=0.0092
p=0.4638
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

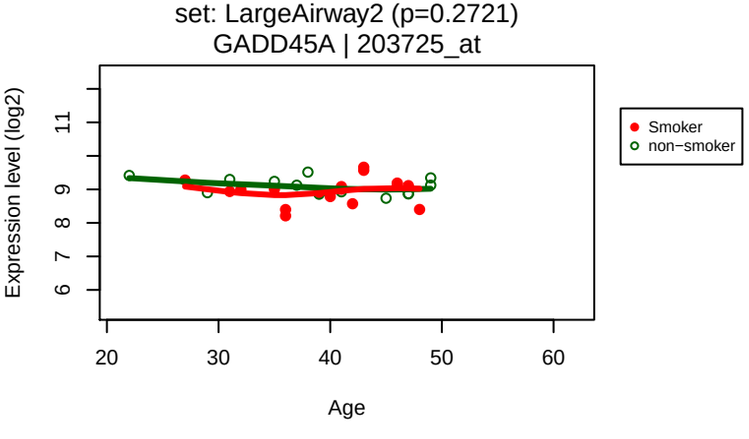

Gene expression ratio (smokers/nonsmokers)
gene: GADD45A | 203725_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0264
p=0.047
p=0.0048
p=0.1911
p=0.2721
p=0.96
p=0.0092
p=0.4638
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

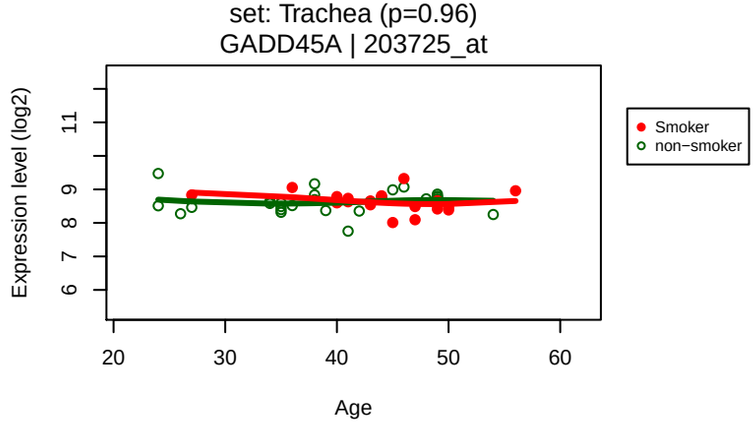

Gene expression ratio in Males
gene: GADD45A | 203725_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2627
p=0.1349
p=0.6104
p=0.4826
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GADD45A | 203725_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0315
p=0.9563
p=0.5309
p=0.7075
SmallAirway2
LargeAirway2
Trachea
Macrophage2

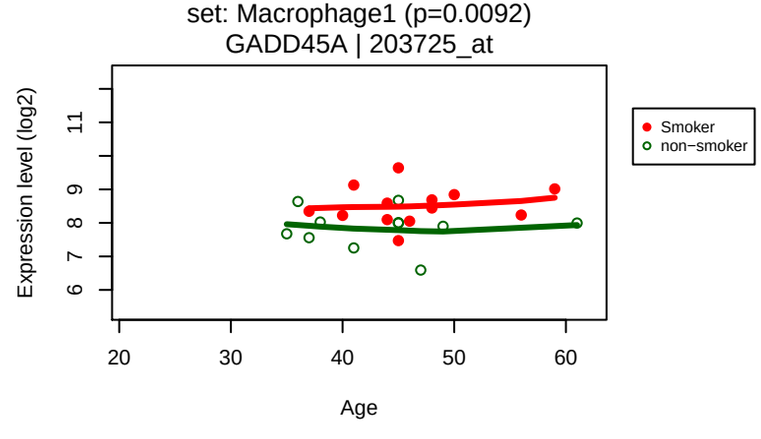

set: SmallAirway1 (p=0.0264)
GADD45A | 203725_at
Expression level (log2)
Age
Smoker
non−smoker

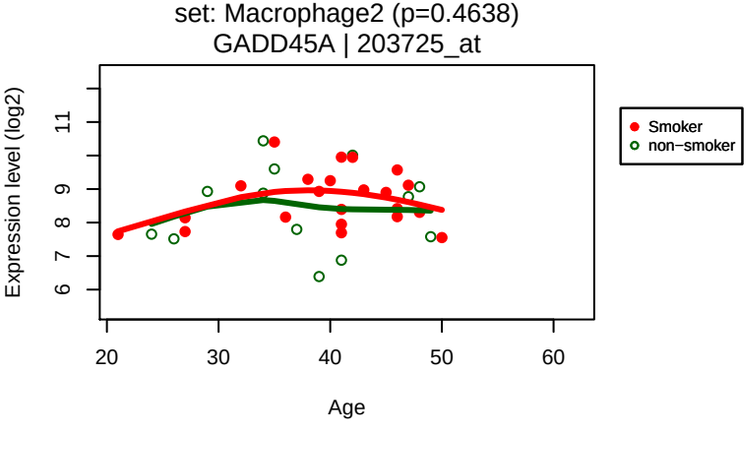

set: SmallAirway2 (p=0.047)
GADD45A | 203725_at
Expression level (log2)
Age
Smoker
non−smoker

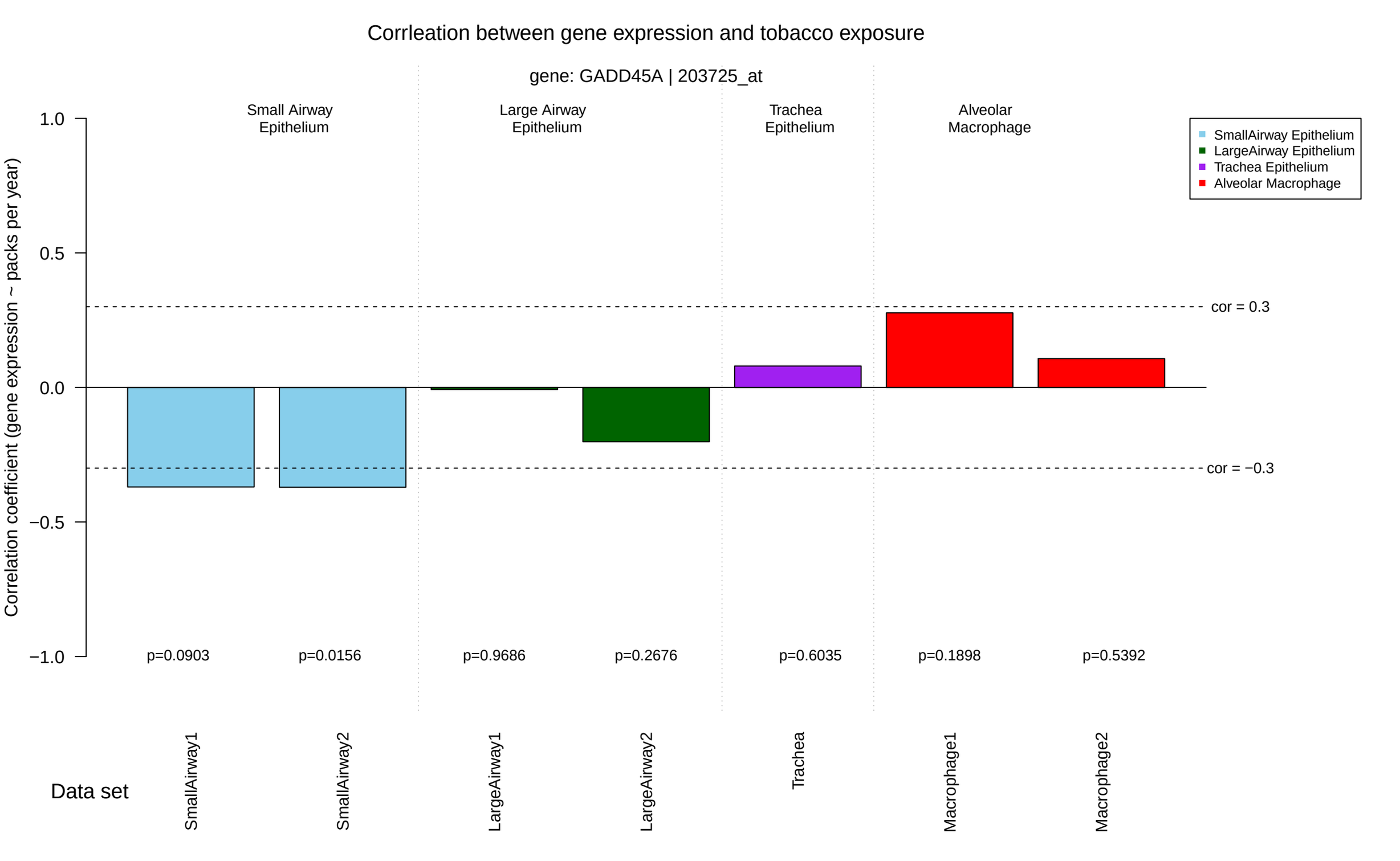

set: LargeAirway1 (p=0.1911)
GADD45A | 203725_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2721)
GADD45A | 203725_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.96)
GADD45A | 203725_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0092)
GADD45A | 203725_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4638)
GADD45A | 203725_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GADD45A | 203725_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0903
p=0.0156
p=0.9686
p=0.2676
p=0.6035
p=0.1898
p=0.5392
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2